Data Structures Notes

Arrays, stacks, queues, trees, graphs, and Big-O

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Arrays

- An array stores items in an ordered sequence, usually accessed by index.
- Reading by index is typically O(1); searching an unsorted array is typically O(n).

Stacks and queues

- A stack is Last In, First Out (LIFO); common operations are push and pop.
- A queue is First In, First Out (FIFO); it is useful for processing tasks in arrival order.

Linked lists

- A linked list stores nodes that refer to other nodes.
- Accessing an arbitrary position usually requires traversal.

Trees and graphs

- A tree organizes nodes hierarchically.
- A graph contains vertices and edges. BFS explores level by level and commonly uses a queue; DFS explores deeply and can use recursion or a stack.

Big-O reminders

- O(1): constant; O(log n): logarithmic; O(n): linear; O(n log n): common efficient sorting; $O(n^2)$: often nested loops.

Practice

- Which structure processes tasks in arrival order?
- What is the typical search time for an unsorted array?
- How do BFS and DFS differ?